1 (a) Write $x^2 - 6x + 1$ in the form $(x + a)^2 + b$ where a and b are integers.

$(x-3)^2 - 9 + 1$

$(x-3)^2 - 8$

$(x-3)^2 - 8$

(2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 - 6x + 1$

$(3, -8)$

(1)

(Total for question 1 is 3 marks)

2 (a) Write $x^2 + 8x + 5$ in the form $(x + a)^2 + b$ where a and b are integers.

$(x+4)^2 - 16 + 5$

$(x+4)^2 - 11$

$(x+4)^2 - 11$

(2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 + 8x + 5$

$(-4, -11)$

(1)

(Total for question 2 is 3 marks)

3 (a) Write $x^2 + 10x + 2$ in the form $(x + a)^2 + b$ where a and b are integers.

$(x+5)^2 - 25 + 2$

$(x+5)^2 - 23$

$(x+5)^2 - 23$

(2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 + 10x + 2$

$(-5, -23)$

(1)

(Total for question 3 is 3 marks)

4 (a) Write $x^2 - 2x - 1$ in the form $(x + a)^2 + b$ where a and b are integers.

$(x-1)^2 - 1 - 1$

$(x-1)^2 - 2$

$(x-1)^2 - 2$

(2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 - 2x - 1$

$(1, -2)$

(1)

(Total for question 4 is 3 marks)

5 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 8x + 3$
You must show all your working.

$(x + 4)^2 - 16 + 3$

$(x + 4)^2 - 13$

$(-4, -13)$

(Total for question 5 is 3 marks)

6 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 10x - 8$
You must show all your working.

$(x + 5)^2 - 25 - 8$

$(x + 5)^2 - 33$

$(-5, -33)$

(Total for question 6 is 3 marks)

7 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 3x - 7$
You must show all your working.

$$\left(x + \frac{3}{2}\right)^2 - \frac{9}{4} - 7$$

$$\left(x + \frac{3}{2}\right)^2 - \frac{9}{4} - \frac{28}{4}$$

$$\left(x + \frac{3}{2}\right)^2 - \frac{37}{4}$$

$\left(-\frac{3}{2}, -\frac{37}{4}\right)$

(Total for question 7 is 3 marks)

8 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 - x + 8$
You must show all your working.

$$\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} + 8$$

$$\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} + \frac{32}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + \frac{31}{4}$$

$\left(\frac{1}{2}, \frac{31}{4}\right)$

(Total for question 8 is 3 marks)

9 (a) Write $2x^2 - 12x + 23$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

$2(x^2 - 6x + 11.5)$

$2((x-3)^2 - 9 + 11.5)$

$2((x-3)^2 + 2.5)$

$2(x-3)^2 + 5$

$2(x-3)^2 + 5$

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 2x^2 - 12x + 23$

$(3, 5)$

(1)

(Total for question 9 is 4 marks)

10 (a) Write $2x^2 + 16x + 26$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

$2(x^2 + 8x + 13)$

$2((x+4)^2 - 16 + 13)$

$2((x+4)^2 - 3)$

$2(x+4)^2 - 6$

$2(x+4)^2 - 6$

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 2x^2 + 16x + 26$

$(-4, -6)$

(1)

(Total for question 10 is 4 marks)

11 (a) Write $3x^2 - 6x + 6$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

$3(x^2 - 2x + 2)$

$3((x-1)^2 - 1 + 2)$

$3((x-1)^2 + 1)$

$3(x-1)^2 + 3$

$3(x-1)^2 + 3$

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 3x^2 - 6x + 6$

$(1, 3)$

(1)

(Total for question 11 is 4 marks)

12 (a) Write $3x^2 - 30x + 63$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

$3(x^2 - 10x + 21)$

$3((x-5)^2 - 25 + 21)$

$3((x-5)^2 - 4)$

$3(x-5)^2 - 12$

$3(x-5)^2 - 12$

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 3x^2 - 30x + 63$

$(5, -12)$

(1)

(Total for question 12 is 4 marks)

13 By completing the square, solve $x^2 + 10x - 3 = 0$
Give your answers in surd form.

$$(x+5)^2 - 25 - 3 = 0$$
$$(x+5)^2 - 28 = 0$$
$$(x+5)^2 = 28$$
$$(x+5) = \pm\sqrt{28}$$
$$x = -5 \pm \sqrt{28}$$
or
$$= -5 \pm 2\sqrt{7}$$

$x = -5+2\sqrt{7}$ or $x = -5-2\sqrt{7}$

(Total for question 13 is 5 marks)

14 By completing the square solve $x^2 + 5x + 4.25 = 0$
Give your answers in surd form.

$$\left(x + \frac{5}{2}\right)^2 - \frac{25}{4} + 4.25 = 0$$
$$\left(x + \frac{5}{2}\right)^2 - 2 = 0$$
$$\left(x + \frac{5}{2}\right)^2 = 2$$
$$x + \frac{5}{2} = \pm\sqrt{2}$$
$$x = -\frac{5}{2} \pm \sqrt{2}$$

$x = -\frac{5}{2} + \sqrt{2}$ or $x = -\frac{5}{2} - \sqrt{2}$

(Total for question 14 is 5 marks)